

Subject Description Form

Subject Code	ME534
Subject Title	Engineering Acoustics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Students should have basic knowledge in Dynamics and Vibrations.
Objectives	To provide the ingredients for students to acquire a sound background in modern acoustics and control of noise.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - possess state-of-the-art knowledge and skills in the area of physical characteristics of sound, noise radiation mechanism and phenomena of sound propagation; - apply their knowledge, skills and hand-on experience to measure and analyse the content of sound and design the noise control system; - extend their knowledge of noise radiation mechanism and noise control principles to different situations of engineering context and professional practice; and - have recognition of the need for, and an ability to engage in life-long learning.
Subject Synopsis/ Indicative Syllabus	Fundamentals of Acoustics: Vibration of single-degree-of-freedom system and continuous system; physical characteristics of sound; noise effect on human beings and noise pollution; human ear; subjective response to noise; wave propagation in fluid media; sound speed; sound energy, power and intensity; modeling of sound waves; Euler's equation of motion; wave equation and Helmholtz equation; general solution to wave equation. Wave Propagation with the Presence of Boundaries: Reflection at rigid and impedance boundaries; sound transmission through interfaces; sound transmission control; reactive silencer; Helmholtz resonator; acoustic modes; calculation of room modes. Noise Analysis: Quantitative measures of sound; frequency content of sounds; acoustic scales; data acquisition and acoustic measurement systems; digital sampling; signal processing; frequency analysis. Noise Sources: Monopole; Dipole; Addition of Sound sources; Sound radiation; Coupling between vibration and sound waves; Elementary sound radiators such as piston and panel; and sound source modeling. Noise Control: Noise attenuation techniques; active noise cancellation; noise barrier and insertion loss; acoustical properties of sound absorbing materials; principles of impedance tube measurement; damping and absorption; viscoelastic damping treatment; impedance of wall structures; calculation of noise level inside a room; transmission and acoustic isolation.

Teaching/Learning Methodology	1. The teaching and learning methods include lectures/tutorial sessions, homework assignments, tests, mini-project or case study, and final examination. 2. The continuous assessment and examination are aimed at providing students with integrated knowledge required for engineering acoustics. 3. Technical/practical examples and problems are raised and discussed in class/tutorial sessions.																																												
	<table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Intended subject learning outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Homework assignment</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Case study report and presentation</td><td>√</td><td>√</td><td>√</td><td></td></tr></table>					Teaching/Learning Methodology	Intended subject learning outcomes				a	b	c	d	1. Lecture	√	√	√	√	2. Tutorial	√	√	√	√	3. Homework assignment	√	√	√	√	4. Case study report and presentation	√	√	√												
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Reading List and References	Textbooks: 1. Hansen C. H. and Snyder S. D., <i>Active Control of Noise and Vibration</i>, Spon, latest eidtion. 2. Pierce A. D., <i>Acoustics</i>, Acoustic Society of America, latest edition. 3. Kleppe J. A., <i>Engineering Application of Acoustics</i>, Artech House, latest edition. 4. Everest F. A., <i>The Master Handbook of Acoustics</i>, Tab Books Inc., latest edition. 5. Bies D. A. and Hansen C. H., <i>Engineering Noise Control</i>, Spon, latest edition. 6. Norton M. P., <i>Fundamentals of Noise and Vibration Analysis for Engineers</i>, Cambridge University Press, latest edition.																																												

	7. Kinsler L. E. et al, <i>Fundamentals of acoustics</i>, Wiley, latest edition. Journals: • The Journal of the Acoustical Society of America, Acoustical Society of America. • Journal of Sound and Vibration, Academic Press. • Acustica united with Acta Acustica, S. Hirzel Verlag. • Applied Acoustics, Elsevier Applied Science.
Last Update	June 2024